

Depot

An infrastructure architect powered by AI,
from code to cloud.

TEAM

Ahsan Riaz

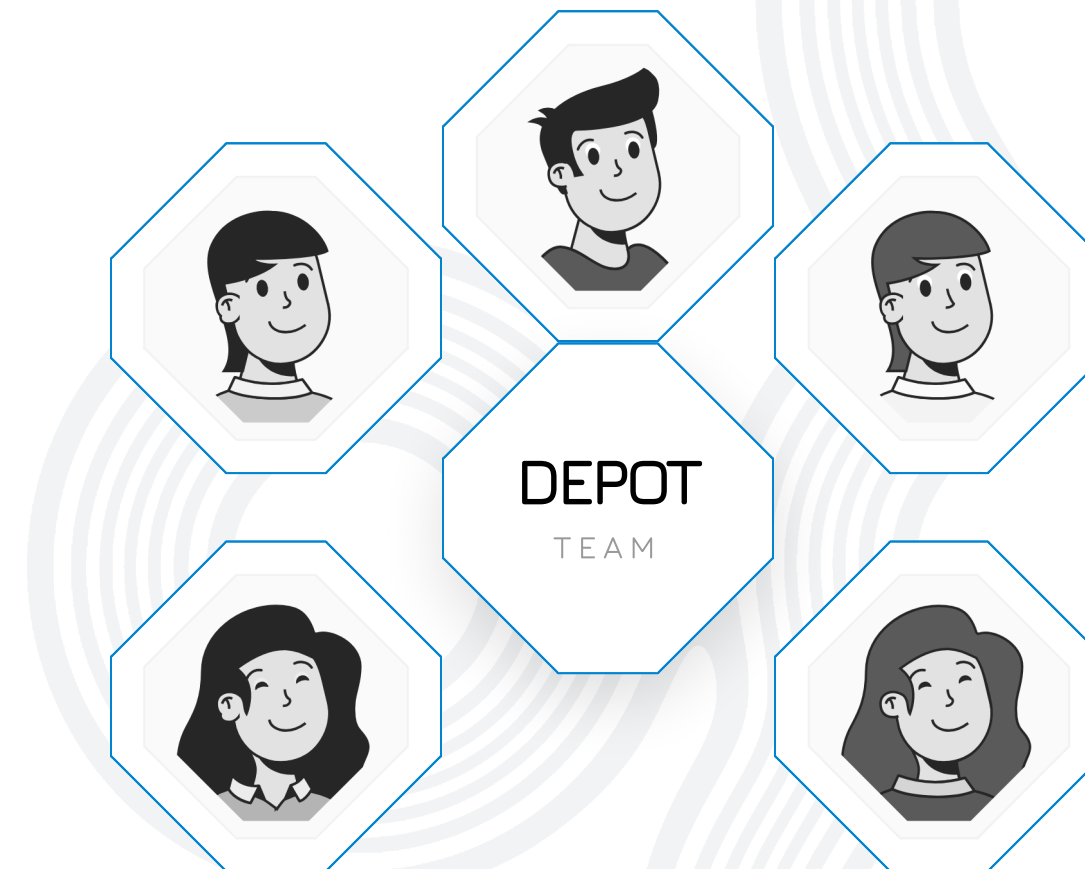
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Building apps is fast now. Shipping them is the bottleneck.

~70%

of AI-written cloud
setups fail on their first
deploy

Even from the best of six AI
models

[Zhang et al., FSE 2026](#) [1]

77%

of AI-written cloud
code that works still
fails a security check

Even from the best model
tested

[Vargas et al., 2026](#) [2]

29%

of company cloud
spend is wasted

Estimated by 753 cloud
decision-makers

[Flexera State of the Cloud, 2026](#) [3]

17.8%

of developers use
Terraform, though
43.3% build on AWS

24,473 developers surveyed

[Stack Overflow Developer Survey,
2025](#) [4]

Working code still leaves four decisions.

01

What it needs

Which servers,
database and storage
the app needs to run.

02

Where it runs

Which cloud company
hosts it: Amazon,
Google or Microsoft.

03

What it costs

The monthly bill,
known before anything
is switched on.

04

How to change it safely

Updating it later
without breaking what
already works.

Depot answers all four, **and explains, records and approves every change.**

Each tool solves part of it.

Someone still decides where it runs.

	Vercel	Porter	Pulumi Neo	Depot
Easy to deploy?	Yes Push code, it's live	Yes Connect a repository	No Needs infrastructure code	Yes Point it at your code
Runs on your own cloud?	No Vercel's platform only	Yes AWS, GCP or Azure	Yes Any cloud	Yes AWS, GCP or Azure
Picks the best-value setup for you?	No Fixed platform	No You choose	No An engineer decides	Yes Compares and prices options

From each product's own documentation [5]. Depot is proposed.

Built for builders without an infrastructure engineer.

AI-assisted builders

A working product, but little
experience of running one.

Solo full-stack developers

Outgrowing a managed
platform: stay, or move to a
cloud?

Small teams and agencies

Two to ten people, with no
DevOps staff.

What Depot does for you.

01

Reads your code

Works out what the app needs from its code and a few questions.

02

Picks the cloud

Compares AWS, Google Cloud and Azure, and chooses the best fit at the lowest cost.

03

Writes the infrastructure

Designs the architecture and writes the Terraform from proven, security-reviewed blueprints.

04

Checks and repairs its work

Validates, scans and prices every change, and fixes failures before you see them.

05

Ships and keeps it running

Deploys on every push, rolls back bad releases and tracks cost.

06

Fixes problems by chat

Diagnoses incidents and proposes fixes and savings for a person to approve.

Three objectives.

What a user can do with Depot.

01

Deploy any supported repository

Code in, live app out, on AWS, GCP or Azure.

02

Check health, metrics and cost

Logs, spend and rollback in one place.

03

Remembers every decision

What changed, when, and why.

Research sits inside all three: comparing ten current AI models on architecture and IaC writing, and measured guardrails for the agent loop.

One session, from working code to live.

Step 1

Connect

GitHub, GitLab or an archive

Point Depot at the app's code, or upload it.



Step 2

Answer

Up to eight questions

Answer simple questions the code can't, like expected traffic and budget.



Step 3

Review

AI-written architecture and Terraform, with monthly cost

See the proposed setup and what it will cost each month.



Step 4

Approve

Signed and role-checked

Nothing goes live until an authorised person signs it off.



Step 5

Run

Deploy on push, roll back, watch cost

The app goes live, updates itself, and can be undone.

How Depot is built: four simple layers.

Layer 1

You

Web dashboard

Chat assistant

Same rules for both: a chat can't do more than a button.



Layer 2

Depot's brain

Reads the code →

Designs and writes the setup →

Checks and repairs it

Drafts from proven blueprints, then tests its own work.



Layer 3

Safety gate

Shows cost and risks

Waits for a signed approval

Nothing reaches the cloud without a person saying yes.



Layer 4

Your cloud

Amazon Web Services

Google Cloud

Microsoft Azure

Runs in the user's own account, with short-lived access.

Alongside every layer

Memory

Every decision, change, cost and approval is recorded, so anyone can see why the setup is what it is.

What Depot covers.

Any web app, API, static site or worker

Written in JavaScript/TypeScript, Python or Go

From GitHub, GitLab or an upload

One code source per app

On AWS, Google Cloud or Azure

In the user's own cloud account

AI-designed infrastructure

Written from proven blueprints, then checked and repaired

Deploys, rollback and monitoring

With cost tracking and savings

A chat assistant for incidents

Diagnoses problems and proposes fixes

Extended scope only if M8 is met early: guided mode, more clouds (e.g. DigitalOcean, Oracle), voice, preview environments.

How we'll prove it works.

Pass marks set before we build.

We measure	Pass if	Context
Deploys successfully	7 of 8 test apps live after repair	Today's AI: 20.8–30.2% on the first try [1]
Picks the right architecture	80% or more accepted by reviewers	Two independent reviewers
Understands the app	85% or more of facts correct	Checked against labelled apps
Estimates the cost	Within 25% of the cloud's own calculator	Before anything is switched on
Cuts the cost	20% or more saved each month	On apps that sit idle
Blocks risky changes	Zero unapproved changes	Every seeded unsafe plan blocked
Easy to use	80% deploy unaided; usability 68 or more [7]	68 is the published average

Pass marks from the FYDP-I proposal, measured in milestone M8.

Three clouds first, then everything else.



Team planning targets; dates will follow the official defence schedule.

The team.

No one reviews only their own work.



Ahsan Riaz

AI & data logic

- Reads code into the Application Model
- Agent mode and incident loop
- Model research and optimisation



Muhammad Waleed

Front end & delivery

- Dashboard and team workspaces
- Deploys, rollback and monitoring
- User research and testing



Muhammad Faizan Anwar

Cloud & backend logic

- AI architect and Terraform writing
- AWS, GCP and Azure adapters
- Safety gate and cost savings

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim**

Two goals we contribute to.

SDG 8.3

Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.

SDG 9.5

Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

Goals and targets from the UN Sustainable Development Goals [6]. This publication has not been approved by the United Nations and does not reflect the views of the United Nations or its officials or Member States.

Sources.

- 1 [T. Zhang, S. Pan, Z. Zhang, Z. Xing and X. Sun, "Deployability-centric infrastructure-as-code generation: Fail, learn, refine, and succeed through LLM-empowered DevOps simulation," ACM FSE, 2026 \(arXiv:2506.05623\).](#)
- 2 [F. L. S. Vargas, R. B. Mansilha and D. Kreutz, "Security-first evaluation of text-to-Terraform: Benchmarking LLMs and SLMs for secure IaC generation," arXiv:2608.02672, 2026.](#)
- 3 [Flexera, 2026 State of the Cloud Report, 2026.](#)
- 4 [Stack Overflow, "Technology," 2025 Stack Overflow Developer Survey, 2025.](#)
- 5 Vendor documentation of the products compared.
- 6 [United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals.](#)
- 7 [J. Sauro, "Measuring usability with the System Usability Scale \(SUS\)," MeasuringU, 2011.](#)

Click a source to open it. Full references are in the proposal.

Thank you. Any questions?

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